INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.C.A.,** Semester: **First**

Stream: **B.C.A.**

Paper Code: **BCACC103**

Paper Name: **Discrete Structure**

Full Marks: **100** Duration: **3 Hours**

Part - A

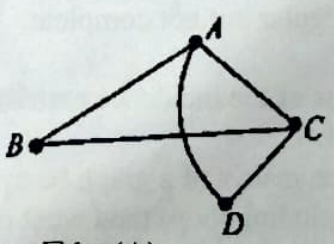
Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

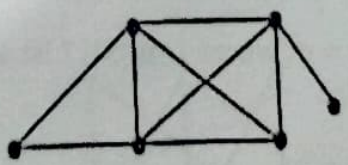
1. Explain the properties of a simple graph with respect to the presence of parallel edges, loops, and isolated vertices. **2**

2. What does the edge-set of a null graph contain? **2**

3. Consider the given graph with vertices A,B,C,D. Identify a circuit in the Graph : **2**



4. For the given graph, calculate ∑deg(A). **2**

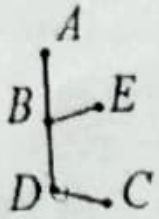


5. What is the maximum degree of any vertex in a simple graph with 10 vertices? 2

6. How many edges has a complete graph with 8 vertices? 2

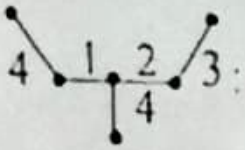
7. What is the minimum possible number of vertices in a tree to be called a binary tree? 2

8. Which one of the vertices of the given tree is the root? 2



9. At what condition a graph G has a spanning tree? 2

10. What is the weight of the given tree? 2



11. Draw a graph which is regular but not complete. 2

12. Identify what the columns of the incidence matrix of a graph represent with respect to the graph itself. 2

13. If the size of the incidence matrix of a graph be m*n and if the graph has 10 vertices and 11 edges (excluding loop) then what is the value of n? 2

14. For the set $\{x : x \in N \text{ and } x = n^2 \text{ where } 1 \leq n \leq 5\}$, List all the elements of the set. 2

15. Let P = {1,2,3}, and Q={2,3,4}. Form the set {(x,y):(x,y)∈P×Q and x≥y}. 2

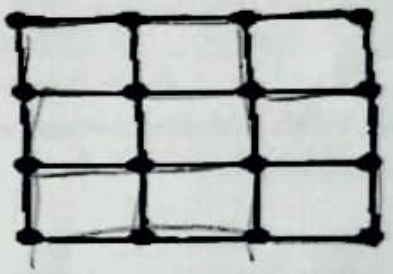
Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Prove that (A×B)∩(C×D)=(A∩C)×(B∩D). 5

17. If A, B and C are any sets, prove that (A-B) U A = A. 5

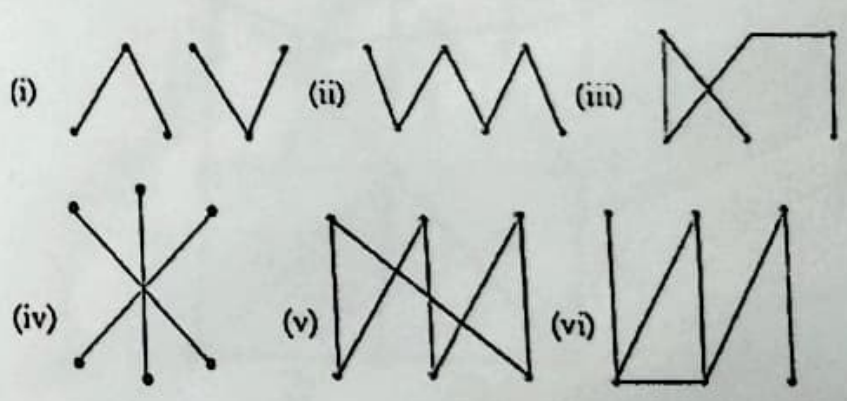
18. Find AU(BΔC) when A={1,2,3,4,5}, B={1,2,4,8}, C={1,2,3,5,7}. 5

19. Find a spanning tree for each of the following graph by removing edges. 5



15

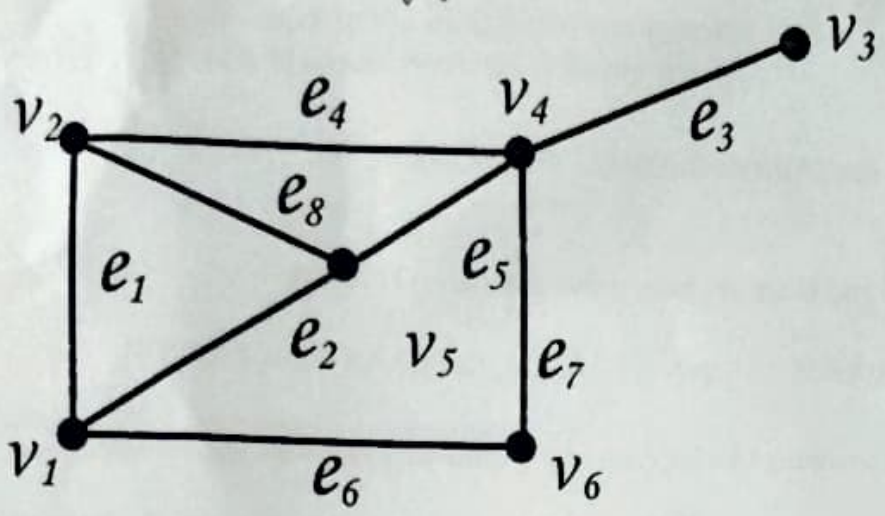
20. Which of the following graphs are tree? 5



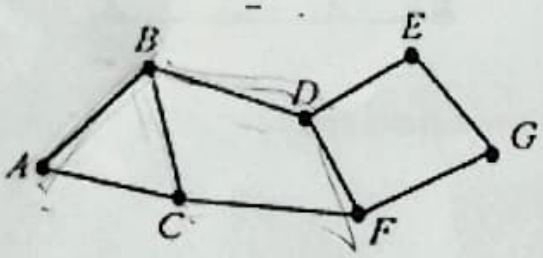
21. Draw a simple graph with five vertices having degree 4, 3, 3, 3, 3. 5

22. Draw all trees with four or fewer vertices. 5

23. Determine the adjacency matrix of the following graph and di-graph: 5



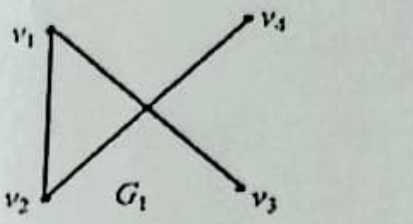
24. In the given graph, construct a path with origin A and terminus C of length 4. Give an another path with same length. 5

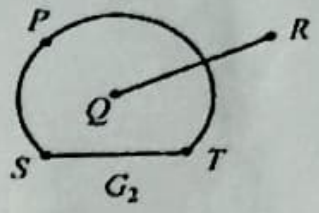


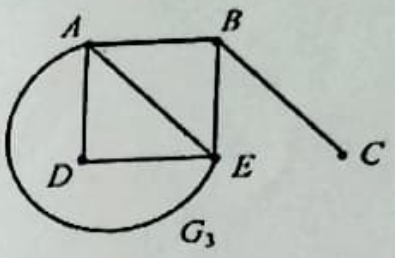
Part - C

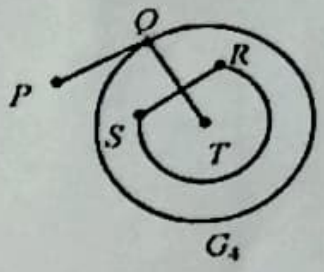
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Four graphs are given below: **10**



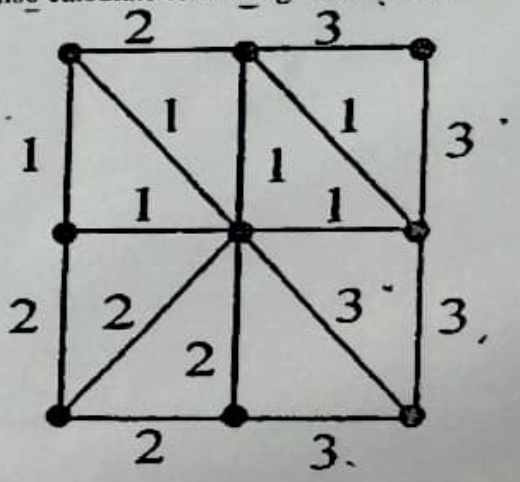






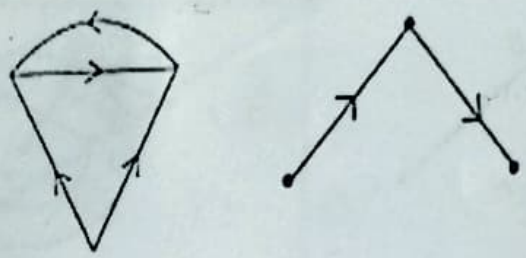
(i) Among those find which one of these are connected? Find the connected components of those which are not connected.
(ii) Find the graphs having no circuit.
(iii) Find the graphs which are loop-free.
(iv) Which are simple graphs?

26. Find by Kruskal's Algorithm, a minimal spanning tree of the following weighted graph: Also calculate total weight of the obtained spanning tree. **10**



27. Draw a tree **10**
(i) with nine edges and nine vertices.
(ii) with six vertices and sum of degrees of all vertices 14.
(iii) with six vertices having degree 1, 1, 1, 1, 3, 3.
(iv) with all vertices of degree 2.

28. Find the Adjacency matrix of the following disconnected graphs and Di-graphs: **10**



29. Let P,Q,R be subsets of a Universal set U. **10**
Prove that
(i) $P\times(Q\cap R)=(P\times Q)\cap(P\times R)$.
(ii) $(P-Q)\times R=(P\times R)-(Q\times R)$.

30. In a survey of 320 persons, number of persons taking tea is 210, taking milk 100 and coffee is 70. Number of persons who take tea and milk is 50, milk and coffee is 30, tea and coffee is 50. The number of persons taking all the three together is 20. Find the number of persons who take neither tea nor milk nor coffee. **10**

31. If U={1,2,3,4,5,6,7,8,9,10}, A={2,5,6,9}, B={1,2,3,4,10}, C={4,6,7,8,9} then **10**
(i) Find A′.
(ii) Verify (A′)′=A.
(iii) Verify De' Morgan's law $(A\cap B)'=A'\cup B'$.
(iv) Verify Distributive law $A\cup(B\cap C)=(A\cup B)\cap(A\cup C)$.

32. a) Find the minimum and maximum number of edges of a simple graph with 10 vertices and 3 components. **10**

b) Draw a connected graph with 10 vertices (i) having 7 edges (ii) having 50 edges.

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